

Elizabeth Jiaxin Zhou

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Summary

Case Western Reserve University Physics and Astronomy Undergraduate and Ohio State University Physics Graduate student with experience in nanophotonics, planetary dynamics, and educational astronomy. Interested in design and manufacture of optics for astronomical instrumentation used in exoplanet detection.

Skills

Data Analysis | Coding: C++, Java, Python, Matlab, Excel, Mathematica | Ansys Zemax Opticstudio | Education | Organization | Communication | Multilingual: English, Chinese

Education

PHYSICS PHD CANDIDATE | AUGUST 2025 - Present | Ohio State University
GPA - 3.5

ADAPTIVE OPTICS SUMMER SCHOOL | AUGUST 2026 | University of California Santa Cruz
PHYSICS BS DEGREE | JANUARY 2022 - MAY 2025 | Case Western Reserve University
GPA - 3.75 | Graduated with Minors in Math and English

ASTRONOMY BS DEGREE | JANUARY 2022 - MAY 2025 | Case Western Reserve University
GPA - 3.75 | Graduated with Minors in Math and English

HIGH SCHOOL DIPLOMA | AUGUST 2017 - MAY 2021 | Cherry Creek High School
GPA - 3.91 | Graduated with High Honors

Research Experience

RESEARCH ASSISTANT | DECEMBER 2025 - Present | Ohio State University
Instrumentation research with focus on designing optics for exoplanet detection.

- Designed radially symmetrical PIAA optic for wavefront apodization
- Simulated PIAA wavefront propagation in both Hcipy and dLux python package

RESEARCH ASSISTANT | DECEMBER 2023 - MAY 2025 | Case Western Reserve University
Planetary dynamics research with a focus on the calculation of external perturbations.

- Derived and verified gravitational asymptotic multipole expansions integrals
- Calculated eigenvectors for dipole, quadrupole, octapole, and hexadecapole systems

RESEARCH ASSISTANT | JANUARY 2023 - MAY 2025 | Case Western Reserve University
Nanophotonics research with a focus on the design and manufacture of metasurfaces.

- Used python package MEEP to simulate nanopillar structures
- Designed and manufactured metasurfaces using the Nanoscribe
- Tested the phase change from infiltrating metasurfaces with liquid crystals

ASTRONOMY INTERN | JUNE 2023 - AUGUST 2023 | Cleveland Museum of Natural History
Educational outreach programming.

- Designed and presented several planetarium shows to public audiences
- Started a monthly astronomy newsletter for the museum members
- Organized a meteor shower viewing event

Work Experience

TEACHING ASSISTANT | AUGUST 2025 - Present | Ohio State University

Educator, lab manager, and grader for undergraduate physics and astronomy courses.
GRADER | JANUARY 2025 - MAY 2025 | Case Western Reserve University
Homework and exam grader for university electromagnetism course.
FACILITY ASSISTANT | MAY 2024 - AUGUST 2024 | Case Western Reserve University
Organization and repair of undergraduate housing facilities.
LIBRARY ASSISTANT | JULY 2023 - JANUARY 2025 | Case Western Reserve University
Organization and assistance of library materials and patrons.
MUSEUM VOLUNTEER | APRIL 2022 - MAY 2025 | Cleveland Museum of Natural History
Astronomy education and outreach.

Honors

COLORADO INDEPENDENT PUBLISHERS ASSOCIATION AWARD | SEPTEMBER 2020
Second place in the action/adventure category of the CIPA awards.
FOUNDING MEMBER OF SIGMA PI SIGMA CHAPTER | NOVEMBER 2023
Founding member of the Case Western Reserve University chapter #591
JAMES C. WYANT AWARD | MAY 2025
Awarded for research accomplishments outside of a senior thesis project.
POLYKARP KUSCH PRIZE | MAY 2025
Awarded for best physics senior thesis.